

POLITECNICO
MILANO 1863

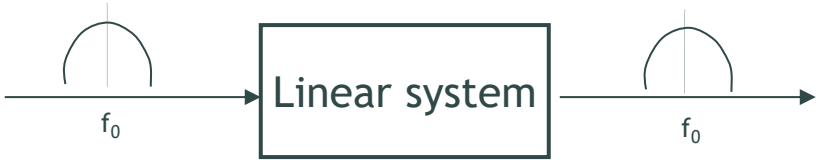
DIPARTIMENTO DI ELETTRONICA
INFORMAZIONE E BIOINGEGNERIA

APPLIED ELECTROMAGNETICS @POLIMI

RF Systems

Non Linearity

RF Systems under large signals excitation



Small signal regime:

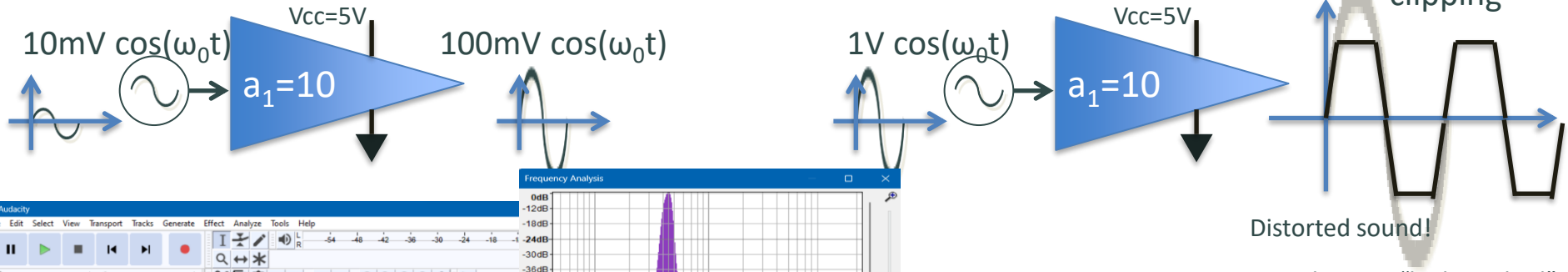
- Everything remains linear
- One tone at ω_0 remains one tone at ω_0, no additional components

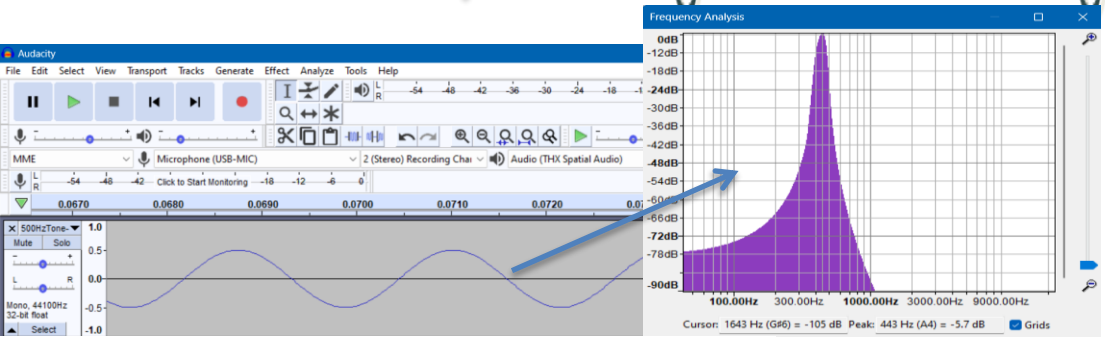
Large signal regime

- non-linear distortion is produced
- increases quickly with the power delivered to the load;
- specs about power must be always accompanied with linearity requirements
- DC→AC conversion: the higher the linearity, the lower the efficiency

Non-linear networks and RF Signals

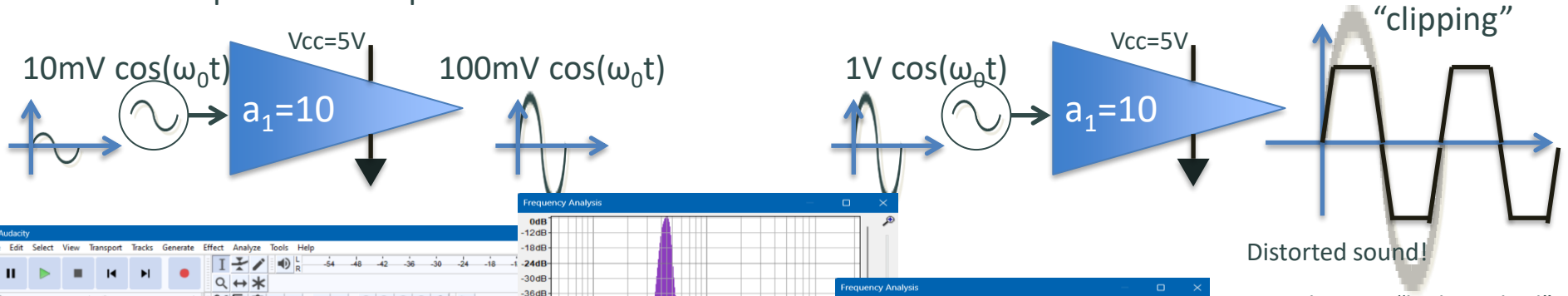
Example: audio amplifier

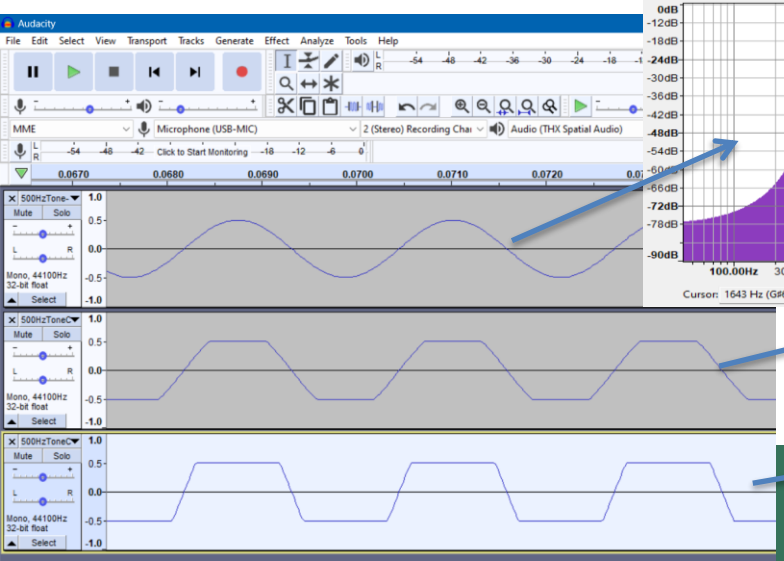


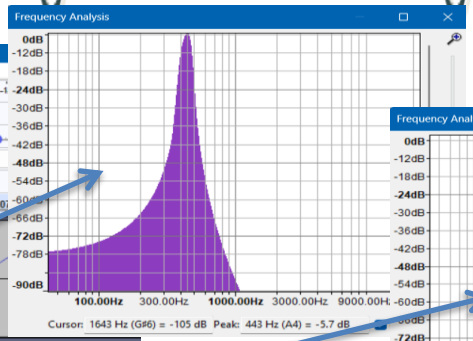


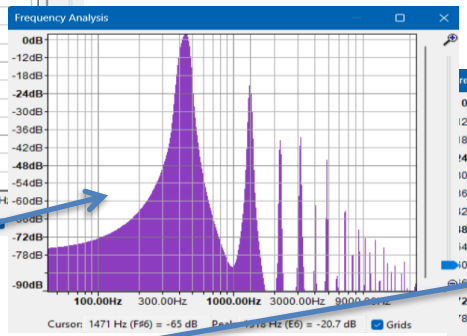
Non-linear networks and RF Signals

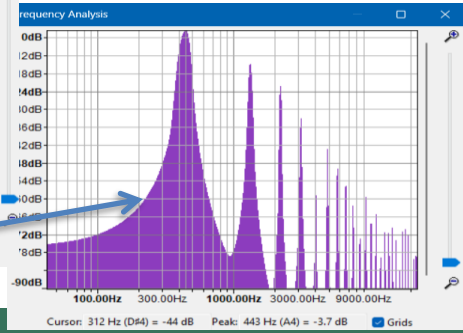
Example: audio amplifier



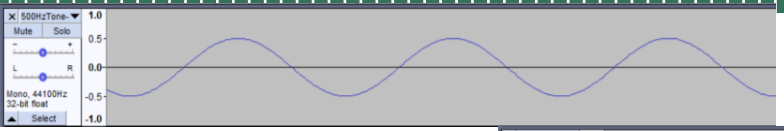


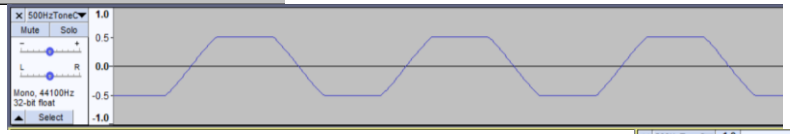


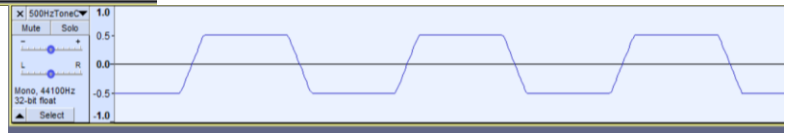




Test







Hear the difference!

1: pure sinusoidal tone

2: moderate clipping

3: heavy clipping

Non-linear networks and RF Signals

In RF: similar behavior + digital receiver more sensitive than human ear

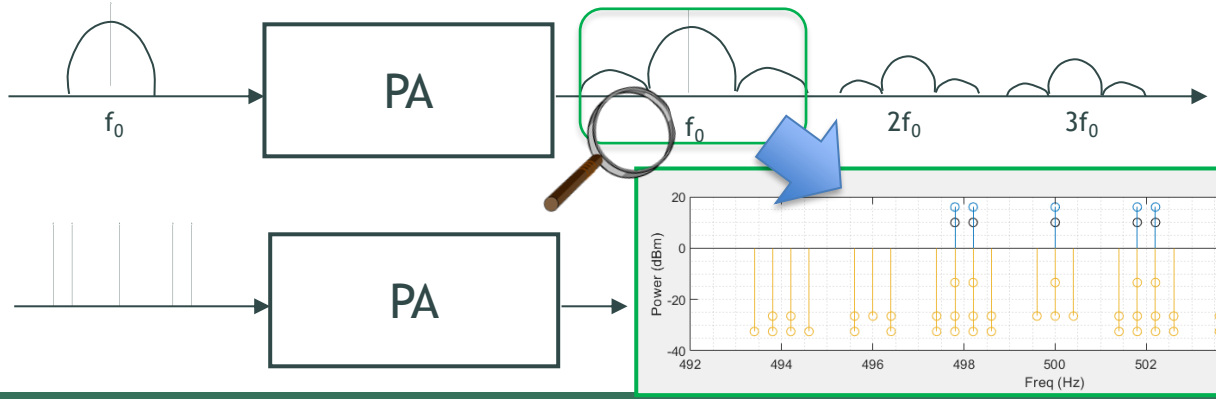
The power amplifier (PA) is usually the one handling the highest RF power, and is thus a very good example to study non-linearity

Non-linear 2-port with an input RF signal:

- Deformation and spreading of the output signal spectrum
- Generation of spurious signals at harmonics of the carrier frequency (f_0)







Some can be filtered out; others cannot

True example of 5 tones into a non-linear amplifier Zoomed to the useful part

Analysis of non-linear network/systems

- Superposition principle: **not applicable** → Analysis in frequency domain & Transfer Funct.: ≈ meaningless.
- A single sinusoidal excitation produces an output with, in general, infinite harmonics

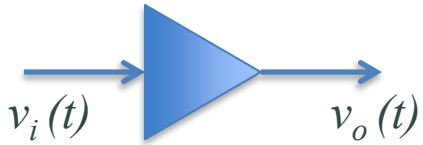


- Simulation: integrating in the time domain the system of differential equations which describe voltages and currents imposed by the components (efficient numerical methods are employed in commercial software like SPICE)
- In case of memoryless networks: an input-output characteristic function can be used for modeling the device

2-port Memoryless Network

Memoryless network: the output voltage at time t_0 depends only on the input voltage at the same time (in networks with memory, instead, it also depends on the instants before).

In this case, the input-output relationship can be analytically described by means of a power series expansion:

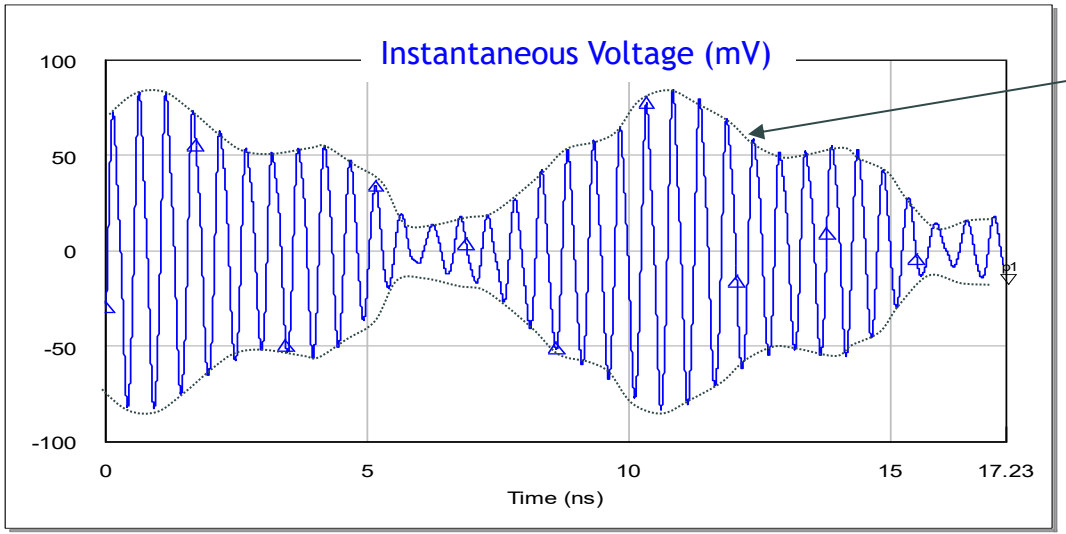


$$v_o = a_0 + a_1 v_i + a_2 v_i^2 + a_3 v_i^3 + \ldots$$

NOTE: The memoryless model is an ideal abstraction. Real networks including reactive components (capacitors, inductors, transmission lines) have always memory.

Peak Envelope Power (PEP)

PEP: “average power of a sinusoidal voltage with amplitude equal to the peak of the modulated signal”
Equivalently: “average power of carrier when modulated at the peak of the envelope”



Envelope Venv(t)

$$PEP \propto \frac{1}{2}|\max(V_{\text{env}}(t))|^2$$

With Z_0 characteristic impedance (usually Z_0=50Ω):

$$PEP = \frac{\left|\max\left(V_{env}(t)\right)\right|^2}{2Z_0}$$

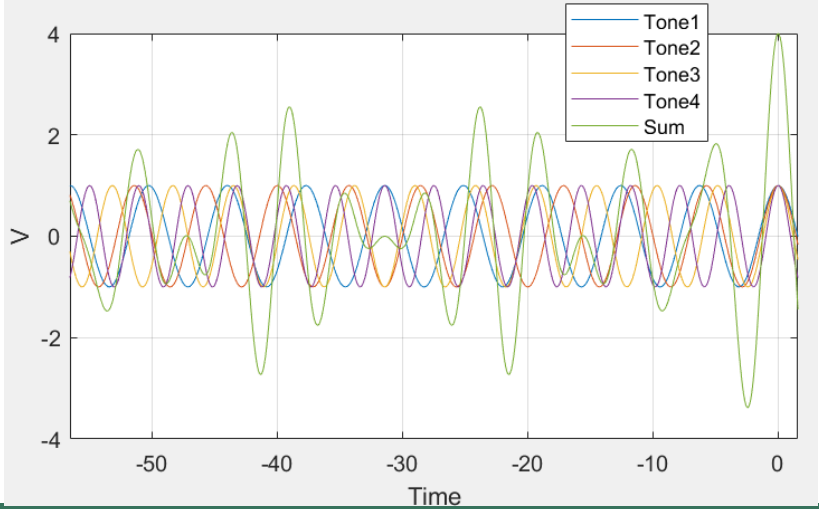
Multi-Tone signal

Signal composed by *n* tones of same amplitude and different frequencies (not harmonically related):

$$V = A\cos(\omega_1 t) + A\cos(\omega_2 t) + A\cos(\omega_3 t) + \ldots + A\cos(\omega_N t)$$

$$P_T = \frac{1}{1/f_i}\int_0^{1/fi} v(i)i(t)dt = f_i \int_0^{1/fi} \frac{(A\cos(2\pi f_i t))^2}{Z_0} dt = \frac{A^2 f_i}{Z_0 2\pi f_i}\int_0^{2\pi} \cos^2(\phi)\, d\phi = \frac{A^2}{Z_0 2\pi}\int_0^{2\pi} (\frac{1}{2} + \frac{1}{2}cos\,(2\phi))d\phi = \frac{A^2}{2Z_0}$$

- A single sinusoid: average power P_T proportional to A^2: $P_T = A^2/(2\,Z_0)$
- Average power: sum of the power of the tones (independent signals): $P_{avg} = N \cdot A^2/(2\,Z_0) = N \cdot P_T$
- Peak power, or Peak Envelope Power (PEP): average power when all sinusoids are at maximum (phasors aligned): $PEP = (N \cdot A)^2/(2Z_0) = N^2 \cdot P_T$
- Peak Factor, Crest Factor, Peak-to Average Ratio: $PF = CF = PAPR = PEP/P_{avg}$ (=N if tones have same amplitude)




```
f1=cos(x); f2=cos(x*1.1); f3=cos(x*1.3); f4=cos(x*1.6); figure, plot(x,[f1;f2;f3;f4; f1+f2+f3+f4]); grid on;
xlabel('Time'); ylabel('V'); legend('Tone1','Tone2','Tone3','Tone4','Sum'); xlim([-9*2*pi,0.25*2*pi]);
```

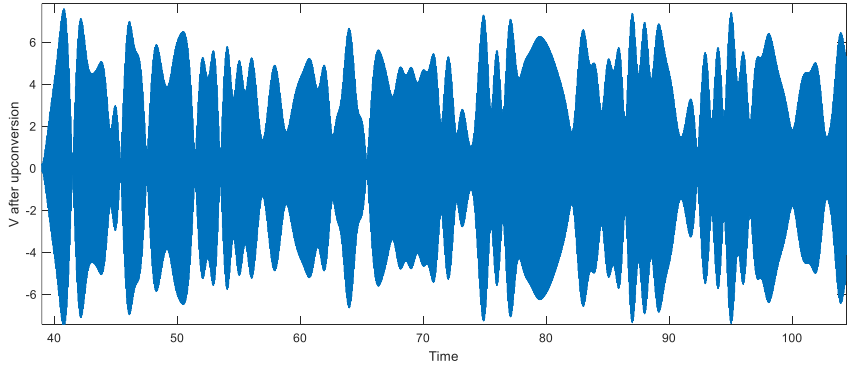
Modulated Signals

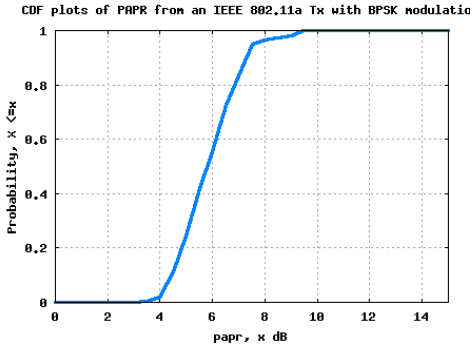
Modulated carriers

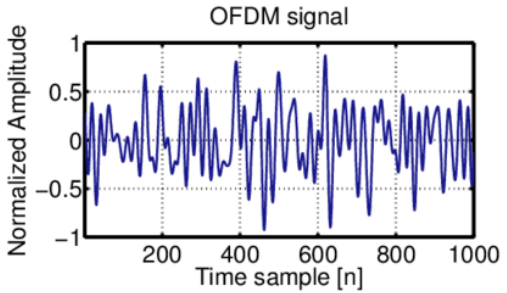
Peak Factor, Crest Factor, Peak-to Average Ratio:

$PF=CF=PAPR=PEP/P_{average}$

- PF depends on modulation & filtering. Statistically defined (Probability to exceed a given value of PF).
- For multicarrier systems (such as OFDM), the value of PF computed as N independent tones is a very rough estimation
- PF is critical to determine the “backoff” with different modulations (see later)

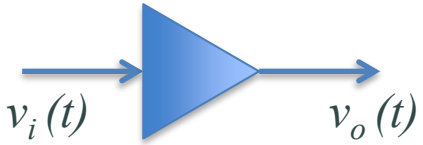






Sinusoidal excitation of a memoryless 2-port

Sinusoids are no longer an exact way to predict arbitrary outputs
But they are still used to characterize non-linear devices



$$v_o = a_1 v_i + a_2 v_i^2 + a_3 v_i^3 +$$

Input: $v_i = A\cos(\omega t)$
Ideal (linear) output: $v_o = a_1 A\cos(\omega t)$
Real Output: $v_o = b_0 + b_1\cos(\omega t) + b_2\cos(2\omega t) + b_3\cos(3\omega t)$

With:

$$v_o = b_0 + b_1\cos(\omega t) + b_2\cos(2\omega t) + b_3\cos(3\omega t) + \ldots$$

$$b_0 = \frac{1}{2}a_2A^2, \qquad b_1 = a_1A + \frac{3}{4}a_3A^3, \qquad b_2 = \frac{1}{2}a_2A^2, \qquad b_3 = \frac{1}{4}a_3A^3$$

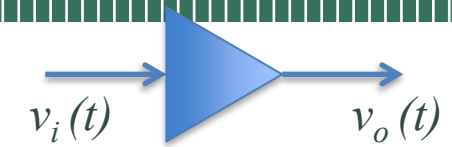
b_1 (amplitude of the output signal at ω) does not depend linearly on A:

there is the non-linear term $(3/4)a_3A^3$

In mplifiers a_3 is normally negative, then there is a **compression** of the output signal.

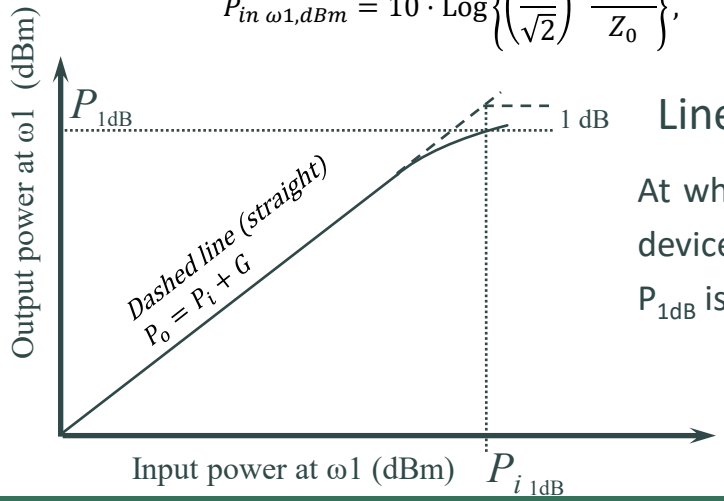
1 dB compression power (P_{1dB})

1-tone test: A sinusoid at frequency f_1 as input
Using memoryless, pure 3rd order model (a_2=0),
input and output power <u>at f_1</u> (the main tone) are:



$$v_o = a_1 v_i + a_3 v_i^3$$

$$P_{in\ \omega 1,dBm} = 10 \cdot \mathrm{Log}\left\{\left(\frac{A}{\sqrt{2}}\right)^2 \frac{1000}{Z_0}\right\}, \qquad P_{o\ \omega 1,dBm} = 10 \cdot \mathrm{Log}\left\{\left(\frac{a_1 A + \left(\frac{3}{4}\right) a_3 A^3}{\sqrt{2}}\right)^2 \frac{1000}{Z_0}\right\}$$



Linear gain (or simply "gain"): $G = 10 \cdot \mathrm{Log}(a_1^2)$

At what power is the output 1dB smaller than what it should be (if the device were linear)? This point is the 1dB compression point.

P_{1dB} is normally measured on the output. ($P_{i\ 1dB}$ is used only for mixers)

With Z_0=50Ω:

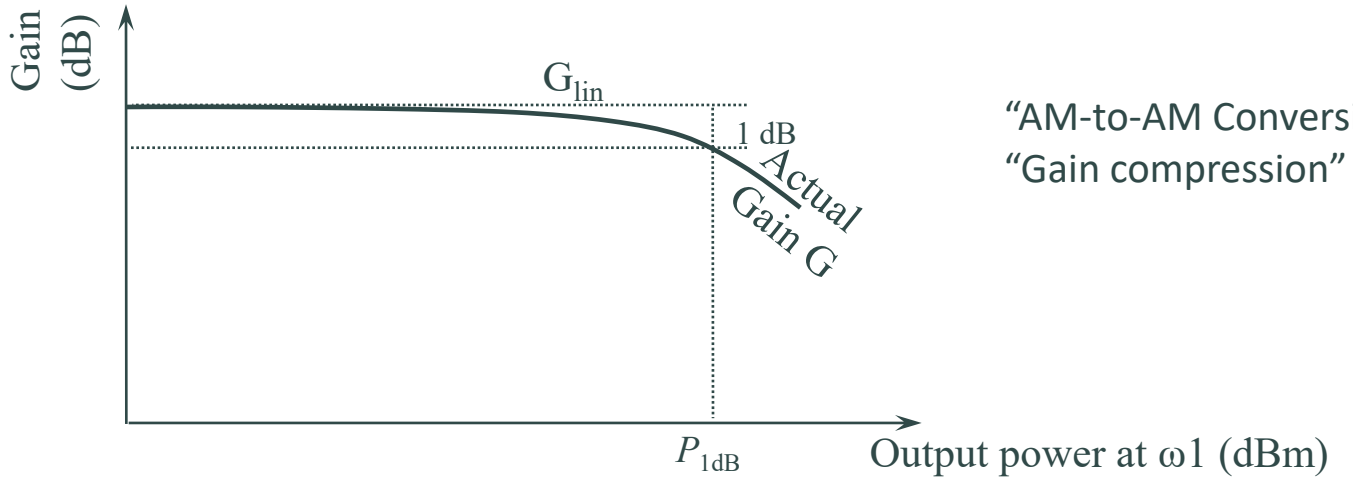
$$P_{1dB} = \left(10\ \mathrm{Log}\frac{a_1^3}{|a_3|} + 0.62\right)\ \mathrm{dBm}$$

Gain vs Output Power

The gain of the PA at the main tone frequency f_1 is a function of A^2:

$$G_{dB} = P_{o\,\omega 1,dBm} - P_{in\,\omega 1,dBm} = 20 \cdot \mathrm{Log}\left\{a_1 + \left(\frac{3}{4}\right) a_3 A^2\right\}$$

G is a function of output power P_o in a 1-tone test:



The nominal gain of an ampli is the gain of the linear term alone:

$G_{lin} = 20 \cdot \mathrm{Log}\{a_1\}$

also known as "linear gain" or "small-signal gain", or often just "gain"

"AM-to-AM Conversion"

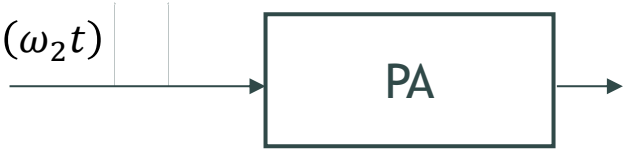
"Gain compression"

Intermodulation by a 3rd order Memoryless Non-Linearity

With a 2-tone input (2-tone test):

$$v_o = a_1 v_i + a_3 v_i^3$$

$$v_i = A\cos(\omega_1 t) + A\cos(\omega_2 t)$$



$$v_o = ?$$

$$a_1 A\cos\omega_1 t + a_1 A\cos\omega_2 t + a_3 A^3(\cos\omega_1 t + \cos\omega_2 t)^3$$

$$\cos^3 x = \frac{3}{4}\cos x + \frac{1}{4}\cos 3x \qquad \cos^2 x = \frac{1}{2} + \frac{1}{2}\cos 2x$$

$$\cos^2 x \cos y = \frac{1}{2}\cos y + \frac{1}{2}\cos 2x \cos y = \frac{1}{2}\cos y + \frac{1}{2}\left(\frac{1}{2}\cos(2x+y) + \frac{1}{2}\cos(2x-y)\right)$$

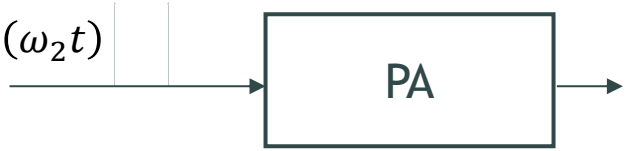
$$\begin{aligned} v_o = {} & a_1 A\cos\omega_1 t + a_1 A\cos\omega_2 t \\ & + a_3 A^3 \frac{3}{4}\cos\omega_1 t + a_3 A^3 \frac{3}{4}\cos\omega_2 t + a_3 A^3 \frac{1}{4}\cos 3\omega_1 t + a_3 A^3 \frac{1}{4}\cos 3\omega_2 t \\ & + \frac{3}{2} a_3 A^3 \cos\omega_1 t + \frac{3}{4} a_3 A^3 \cos(2\omega_1 - \omega_2)\, t + \frac{3}{4} a_3 A^3 \cos(2\omega_1 + \omega_2)\, t \\ & + \frac{3}{2} a_3 A^3 \cos\omega_2 t + \frac{3}{4} a_3 A^3 \cos(2\omega_2 - \omega_1)\, t + \frac{3}{4} a_3 A^3 \cos(2\omega_2 + \omega_1)\, t \end{aligned}$$

Intermodulation by a 3rd order Memoryless Non-Linearity

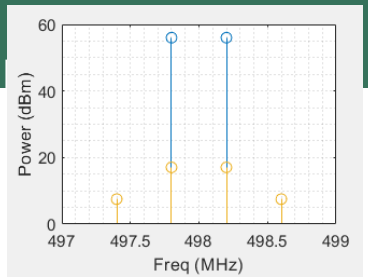
With a 2-tone input (2-tone test):

$$v_o = a_1 v_i + a_3 v_i^3$$

$$v_i = A\cos(\omega_1 t) + A\cos(\omega_2 t)$$



$$v_o = a_1 A\cos\omega_1 t + a_1 A\cos\omega_2 t + a_3 A^3(\cos\omega_1 t + \cos\omega_2 t)^3$$



$$\cos^3 x = \frac{3}{4}\cos x + \frac{1}{4}\cos 3x \qquad \cos^2 x = \frac{1}{2} + \frac{1}{2}\cos 2x$$

$$\cos^2 x \cos y = \frac{1}{2}\cos y + \frac{1}{2}\cos 2x \cos y = \frac{1}{2}\cos y + \frac{1}{2}\left(\frac{1}{2}\cos(2x+y) + \frac{1}{2}\cos(2x-y)\right)$$

$$\begin{aligned} v_o &= a_1 A\cos\omega_1 t + a_1 A\cos\omega_2 t \\ &+ a_3A^3\frac{3}{4}\cos\omega_1 t + a_3A^3\frac{3}{4}\cos\omega_2 t + a_3A^3\frac{1}{4}\cos 3\omega_1 t + a_3A^3\frac{1}{4}\cos 3\omega_2 t \\ &+ \frac{3}{2}a_3A^3\cos\omega_1 t + \frac{3}{4}a_3A^3\cos(2\omega_1-\omega_2)t + \frac{3}{4}a_3A^3\cos(2\omega_1+\omega_2)t \\ &+ \frac{3}{2}a_3A^3\cos\omega_2 t + \frac{3}{4}a_3A^3\cos(2\omega_2-\omega_1)t + \frac{3}{4}a_3A^3\cos(2\omega_2+\omega_1)t \end{aligned}$$

$$\begin{aligned} v_o &= \left(a_1A + \frac{3}{4}a_3A^3 + \frac{3}{2}a_3A^3\right)\cos\omega_1 t \\ &+ \left(a_1A + \frac{3}{4}a_3A^3 + \frac{3}{2}a_3A^3\right)\cos\omega_2 t \\ &+ \frac{3}{4}a_3A^3\cos(2\omega_1-\omega_2)t + \frac{3}{4}a_3A^3\cos(2\omega_2-\omega_1)t \\ &+ a_3A^3\frac{1}{4}\cos 3\omega_1 t + a_3A^3\frac{1}{4}\cos 3\omega_2 t \\ &+ \frac{3}{4}a_3A^3\cos(2\omega_1+\omega_2)t + \frac{3}{4}a_3A^3\cos(2\omega_2+\omega_1)t \end{aligned}$$

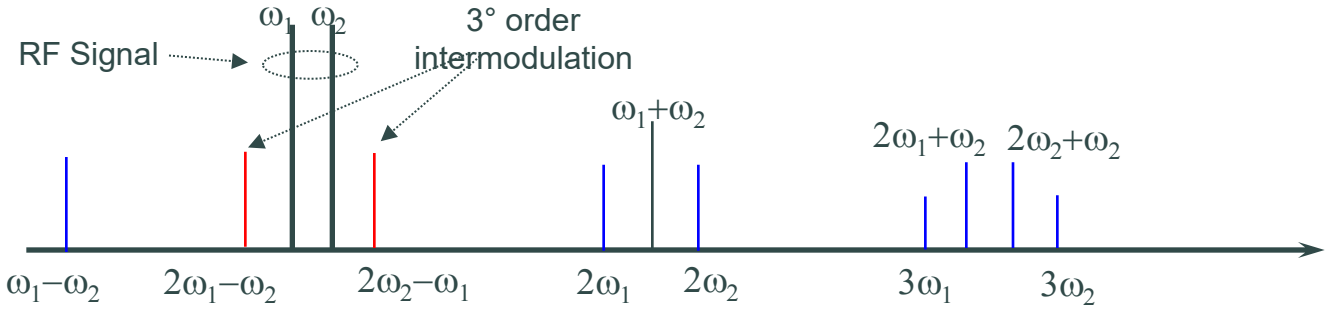
Two-tone Excitation

When a non-linear device is excited by several tones, **intermodulation** is produced:

- in addition to the harmonics of each tone, new tones at the frequencies $|\pm n\omega_1 \pm m\omega_2 \pm k\omega_3 \pm \ldots|$ are output
- In case of two tones with the same amplitude A and frequencies f_1 and f_2, the following tones are generated by a 3rd order non-linearity: $v_o = a_1 v_i + a_2 v_i^2 + a_3 v_i^3$

Frequency	Order	Expression
f_1, f_2	1	$a_1A+(9/4)a_3A^3$
$(f_1-f_2), (f_1+f_2)$	2	a_2A^2
$(2f_1-f_2), (2f_2-f_1)$	3	$(3/4)a_3A^3$
$(2f_1+f_2), (2f_2+f_1)$	3	$(3/4)a_3A^3$
$(2f_1, 2f_2)$	2	$(1/2)a_2A^2$
$(3f_1, 3f_2)$	3	$(1/4)a_3A^3$

Spectrum with two tones



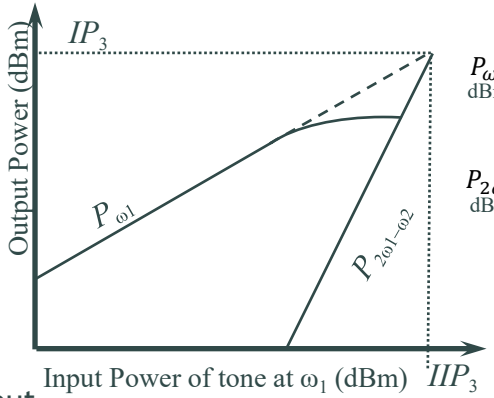
Two tones very close in frequency is the simplest model of a RF signal (the envelope is periodic with period $2/\Delta f$).

Intermodulation products close to 0, $2\omega_1$, $3\omega_1$,… can be easily filtered out

Tones closest to the RF signal cannot be filtered out

- At ω1 and ω2: the useful tones are compressed (more than with 1 tone)
- At $(2\omega_1-\omega_2)$, $(2\omega_2-\omega_1)$: intermodulation products visible nearby the useful signal
  their amplitude is an evident sign of the non-linearity of the amplifier

Intercept Point (IP_3) or Third Order Intercept (TOI)



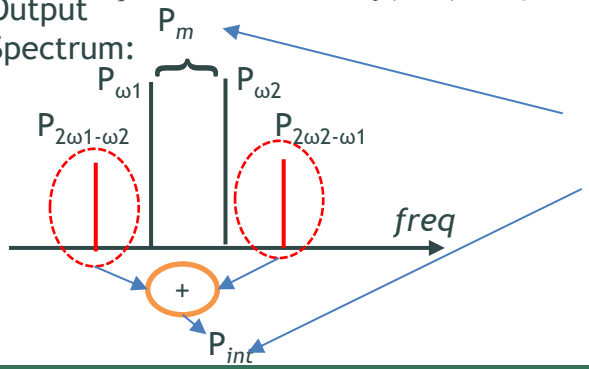
$$P_{\underset{\text{dBm}}{\omega_1}} = 10 \cdot \text{Log}\left\{\left(\frac{a_1 A + \left(\frac{9}{4}\right) a_3 A^3}{\sqrt{2}}\right)^2 \frac{1000}{Z_0}\right\} \cong 10 \cdot \text{Log}\left\{\left(\frac{a_1 A}{\sqrt{2}}\right)^2 \frac{1000}{Z_0}\right\} \quad \text{(neglect compression of main tone)}$$

$$P_{\underset{\text{dBm}}{2\omega_1-\omega_2}} = 10 \cdot \text{Log}\left\{\left(\frac{\left(\frac{3}{4}\right) a_3 A^3}{\sqrt{2}}\right)^2 \frac{1000}{Z_0}\right\} \rightarrow \quad IP_{\underset{\text{dBm}}{3}} = TOI_{\underset{\text{dBm}}{}} = 10 \cdot \text{Log}\left\{\left(\frac{2}{3}\frac{a_1^3}{|a_3|}\right)\frac{1000}{Z_0}\right\} = \left(10 \cdot \text{Log}\frac{a_1^3}{|a_3|} + 11.25\right)_{\text{dBm}}$$

For $Z_0 = 50\Omega$

IP3, or TOI is the output power of the tone at ω_1 where Intercept occurs
IIP3 is the corresponding input power; IIP3 is used only for mixers. $IIP_{3_{dBm}} = IP_{3_{dBm}} - G_{dB}$

Output Spectrum:



Assuming linear $P_{\omega 1}$ (neglect its compression), each intermodulation product is:

$$P_{\underset{\text{dBm}}{2\omega_2-\omega_1}} \simeq 3P_{\underset{\text{dBm}}{\omega_1}} - 2IP_{3_{\text{dBm}}}$$

$$P_{2\omega_2-\omega_{1_{dBm}}} \approx 3P_{\omega 1_{dBm}} - 2IIP_{3_{dBm}} - 2G_{dB}$$

$P_m = (P_{\omega 1} + 3\text{dB})$ the average power of the output two-tone signal
$P_{int} = (P_{2\omega 2-\omega 1} + 3\text{dB})$ is the total mean power in the intermodulation lines (together). Replacing in the above equation:

$$P_{int_{dBm}} \approx 3P_{m_{dBm}} - 2IP_{3_{dBm}} - 6_{dB}$$

Relationship between P_{1dB} e IP_3

IP_3 and P_{1dB} can be expressed, with Z_0=50 Ohm, as:

$$IP_{3\,\mathrm{dBm}} = \left(10 \cdot \mathrm{Log}\frac{a_1^3}{|a_3|} + 11.25\right)_{\mathrm{dBm}}, \qquad P_{1dB\,\mathrm{dBm}} = (10\,\mathrm{Log}\frac{a_1^3}{|a_3|} + 0.62)_{\mathrm{dBm}}$$

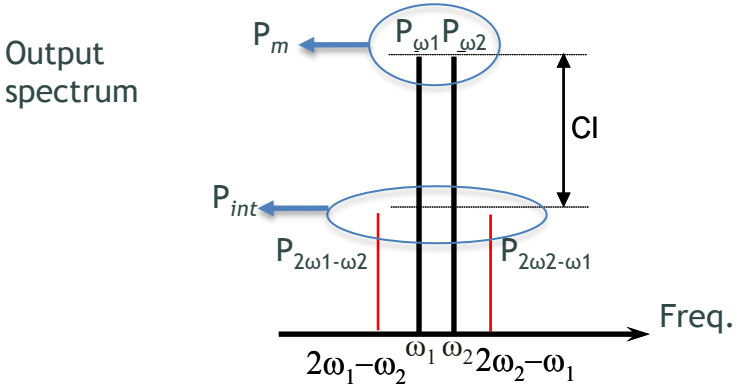
Combining the two relations:

$$IP_{3\,\mathrm{dBm}} = P_{1dB\,\mathrm{dBm}} + \Delta_{p\ \mathrm{dB}} \quad (\Delta_p = 10.63\ \mathrm{dB})$$

The value of Δ_p (10.63dB) refers to an ideal device with a pure 3th order non-linearity.
Real devices may have different Δ_p :

- usually in the range 9-11 dB
- when the transistor is operated very close to its maximum operating frequency, it can be as low as 4-5dB

Carrier-to-intermodulation ratio

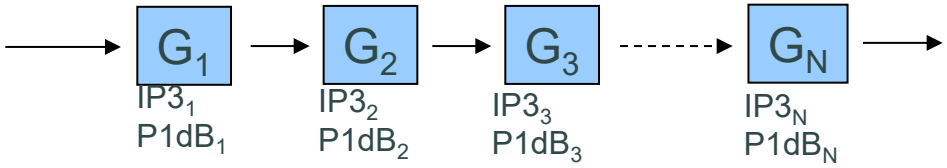


CI: ratio between the main tones power and intermodulation power:

$$CI_{dB} = \underset{\text{dBm}}{P_{\omega_1}} - \underset{\text{dBm}}{P_{2\omega_2-\omega_1}} = \underset{\text{dBm}}{2IP_3} - \underset{\text{dBm}}{2P_{\omega_1}} = \underset{\text{dBm}}{2IP_3} - 2(\underset{\text{dBm}}{P_m} - 3dB) = \underset{\text{dBm}}{2IP_3} - \underset{\text{dBm}}{2P_m} + 6dB$$

$$CI_{dB} \approx \underset{\text{dBm}}{2IP_3} - \underset{\text{dBm}}{2P_m} + 6dB$$

P_{1dB} and IP3 of cascaded 2-port non-linear devices



These formulas are obtained by summing all the distortion <u>voltages</u> produced by the stages [WORST CASE, CONSERVATIVE]

$$\left(\frac{1}{ip3}\right) = \left(\frac{1}{g_2 g_3 \cdots g_N ip3_1}\right) + \left(\frac{1}{g_3 g_4 \cdots g_N ip3_2}\right) + \left(\frac{1}{g_4 g_5 \cdots g_N ip3_3}\right) + \ldots \left(\frac{1}{ip3_N}\right)$$

$$\left(\frac{1}{p_{1dB}}\right) = \left(\frac{1}{g_2 g_3 \cdots g_N\, p1dB_1}\right) + \left(\frac{1}{g_3 g_4 \cdots g_N\, p1dB_2}\right) + \left(\frac{1}{g_4 g_5 \cdots g_N\, p1dB_3}\right) + \ldots \left(\frac{1}{p1dB_N}\right)$$

All in linear values (not in dBm)!

In practice (i.e. with memory effects distortion components tend to sum up in power) [MORE PRACTICAL APPROXIMATION]

$$\left(\frac{1}{ip3}\right)^2 = \left(\frac{1}{g_2 g_3 \cdots g_N ip3_1}\right)^2 + \left(\frac{1}{g_3 g_4 \cdots g_N ip3_2}\right)^2 + \left(\frac{1}{g_4 g_5 \cdots g_N ip3_3}\right)^2 + \ldots \left(\frac{1}{ip3_N}\right)^2$$

$$\left(\frac{1}{p_{1dB}}\right)^2 = \left(\frac{1}{g_2 g_3 \cdots g_N\, p1dB_1}\right)^2 + \left(\frac{1}{g_3 g_4 \cdots g_N\, p1dB_2}\right)^2 + \left(\frac{1}{g_4 g_5 \cdots g_N\, p1dB_3}\right)^2 + \ldots \left(\frac{1}{p1dB_N}\right)^2$$

All in linear values (not in dBm)!

The overall IP3 and P1dB are normally dominated (limited) by the LAST STAGE!
The previous stages only introduce a further small degradation

Tools available!
https://www.analog.com/en/resources/design-tools-and-calculators/rf-and-synthesis-tools.html
& Cadence AWR Design Environment & Keysight SystemVue

“Power is nothing without control”

Any “power” figure of an ampli should come with the corresponding level of nonlinear distortion.
An amplifier datasheet might provide the maximum output PEP power for which the **CI is equal to 30 dB in a 2-tone test**:

$$CI_{(dB)} = 2IP3_{(dBm)} - 2P_{m_{(dBm)}} + 6dB; \qquad PEP_{(dBm)} = P_{m_{(dBm)}} + 3\text{dB} \quad (2-\text{tone signal}),$$

$$30\text{dB} = 2\text{IP3}_{(\text{dBm})} - 2\left(PEP_{(dBm)} - 3\text{dB}\right) + 6\text{dB} \Rightarrow PEP_{(dBm)\,@CI=30dB} = IP3_{(dBm)} - 9dB$$

IP_3 and P_{1dB} are related by:

$$IP3_{(dBm)} = P1dB_{(dBm)} + \Delta p_{(dB)}$$

Then:

$$PEP_{(dBm)@CI=30dB} = P1dB_{(dBm)} + (\Delta_{P_{dB}} - 9dB)$$

Special case: In case Δ_p is ~9 dB, $PEP \approx P_{1dB}$ for CI=30 dB

Back-off

Back-off: $BO_{dB} = P_{1dB,dBm} - AvgPowerOfUsefulSignalAtOutput$

(difference in dB between P_{1dB} and the avg power of the desired "useful" components at output)

- <u>BO is not a "property" of the amplifier itself:</u> BO is the condition at which the amplifier is operating
  - Operating close to P1dB (low BO): the amplifier is working at its highest power, but with severe non-linear distortions
  - Operating far from P1dB (high BO): the amplifier is working at a low power, hence producing smaller non-linear distortion
- In 1-tone test: $BO_{dB} = P_{1dB,dBm} - P_{o,\omega 1}$
- In 2-tone test: $BO_{dB} = P_{1dB,dBm} - P_m$

$$CI_{dB} = 2IP_{3,dBm} - 2P_{m,dBm} + 6dB = 2P_{1dB,dBm} + 2\Delta_{p,dB} - 2P_{m,dBm} + 6dB = 2\left(P_{1dB} + \Delta_p - P_m + 3\right)$$

$$\rightarrow P_{1dB,dBm} = \frac{CI_{dB}}{2} - \Delta_{p,dB} + P_{m,dBm} - 3dB \Rightarrow BO_{dB} = P_{1dB,dBm} - P_{m,dBm}$$

$$\boxed{BO_{dB} = \frac{CI_{dB}}{2} - \Delta_{p,dB} - 3dB}$$

When the output signal is at $PEP_{C/I\ 30dB}$, in a pure 3rd order device, then:

CI=30 dB and Δ_p=10.63dB → BO≈1.4 dB.

<u>NOTE:</u> In other practical cases, we may have Δ_p = 9 dB → To maintain CI=30dB we find BO ≈ 3 dB

Summary of the non-linearity concepts and formulas

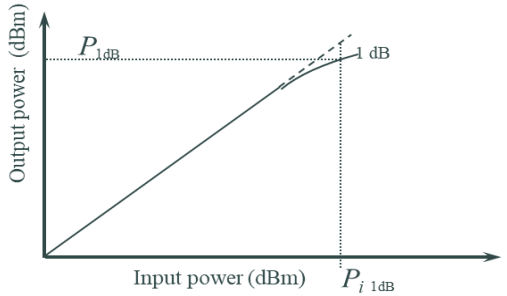
1-tone test

- Gain compression → P_{1dB} (normally at output); At input: P_{i1dB} (dBm) = P1dB (dBm) + 1dB − G (dB)
- Harmonics at $n \cdot f_0$ → Easily filtered out

2-tone test

- Harmonics → Easily filtered out
- Intermod Products at $|n \cdot f_1 \pm m \cdot f_2|$
  - Most dangerous those at $||n|-|m||=1$: in band
- 3rd order non-linearity:
  - Intercept point IP3, TOI (this output power is never reached in practice)
  - Intermod products grow with $3 \cdot Pm_{dB}$:
- C/I ratio: power of amplified tones/power of undesired intermod products
- Often an amplifier is characterized by its output PEP power to guarantee C/I=30dB
- Backoff: used to measure "how far" the actual output power is from the P1dB
- Normally the amplifier operates below P1dB (BO>0dB) otherwise distortion is excessive

$$IP_{3\,\mathrm{dBm}} = P_{1dB\,\mathrm{dBm}} + \Delta_{p\,\mathrm{dB}} \qquad (\Delta_{p\,dB}=9\text{-}11dB) \quad (\Delta_{p\,\mathrm{dB}} = 10.63\ \mathrm{dB})$$



$$P_{2\omega_2-\omega_1\,\mathrm{dBm}} \simeq 3P_{\omega_1\,\mathrm{dBm}} - 2IP_{3\,\mathrm{dBm}}$$

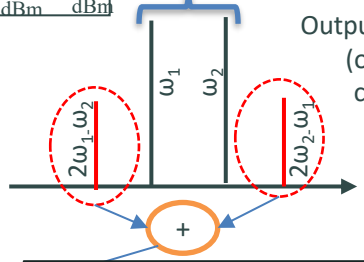
$$P_{m\ (dBm)} = (P_{\omega 1\ (dBm)} + 3dB)$$

$$CI_{dB} \simeq 2IP_{3\,\mathrm{dBm}} - 2P_{m\,\mathrm{dBm}} + 6_{\mathrm{dB}}$$

$$PEP_{\mathrm{C/I\ 30dB\ dBm}} = P_{1dB\,\mathrm{dBm}} + (\Delta_{P\,\mathrm{dB}} - 9_{\mathrm{dB}})$$

$$BO_{\mathrm{dB}} = \frac{CI_{\mathrm{dB}}}{2} - \Delta_{p\,\mathrm{dB}} - 3$$

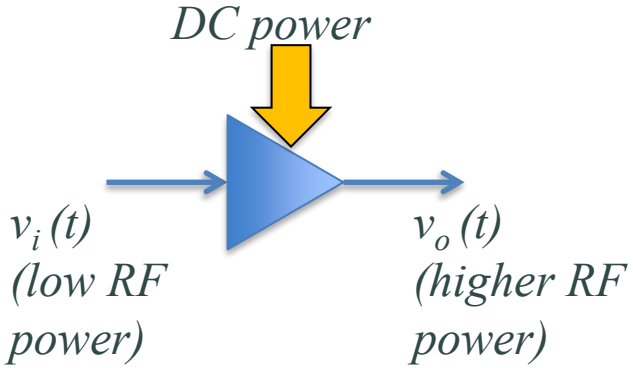
Output spectrum (only in-band components shown)



$$P_{int\ dBm} \approx P_{2\omega_2-\omega_1,dBm} + 6_{\mathrm{dB}} = 3P_{m,dBm} - 2\mathrm{IP}_{3\ dBm} - 6_{\mathrm{dB}}$$

DC-AC Efficiency Conversion

An amplifier is a kind of converter of electric energy from DC to RF.



Power Added Efficiency (PAE):

$$\text{PAE} = 100\frac{(\text{RF Power at load} - \text{RF Power at input})}{\text{DC Power}} \quad \%$$

DC power is delivered by the biasing sources of PA (mostly the drain bias current).

PAE is strictly related to the operating class of the PA:

Increasing linearity, PAE decreases
Maximum linearity, lowest efficiency: class A, AB
Worse linearity, highest efficiency: class E, F, G

Real world signals

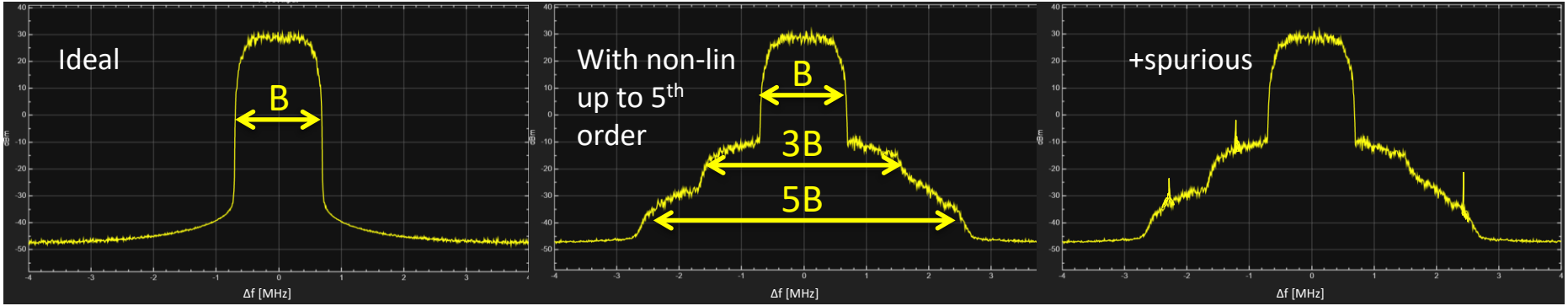
2-tone test: ok to characterize non-linearity; but cannot predict distortion of an arbitrary signal (e.g. IMD)!

For a given average output signal power, intermodulation distortion (IMD) increases with PF, e.g. in a multi-tone signal (high PF means there are high peaks rather than uniform amplitude, and peaks will be strongly deformed)

Input signal is in general defined by its spectrum, assumed limited to band B

Distortion → New frequency components, both in the band of the signal and outside the band

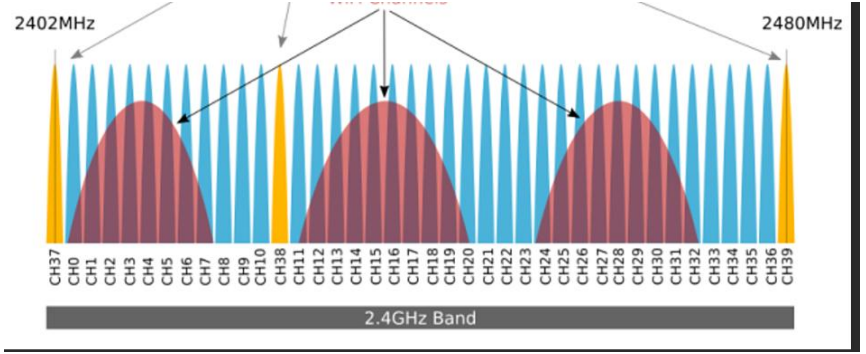
Other spurious signals (e.g. local oscillator leakage) may also be present in the output



→ Further figures of merit have been introduced

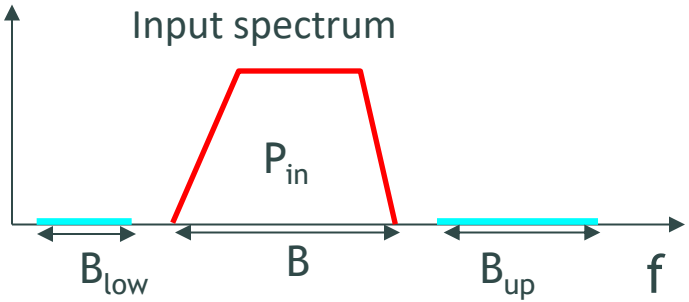
Allocated Spectrum

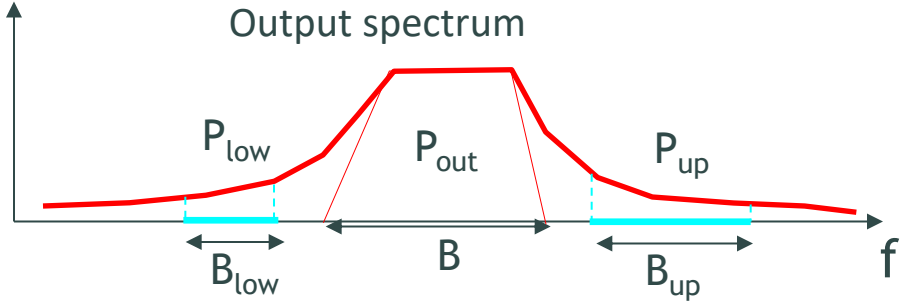




Coexistence of many systems, each with many channels is hard!

Adjacent channel power ratio (ACPR)





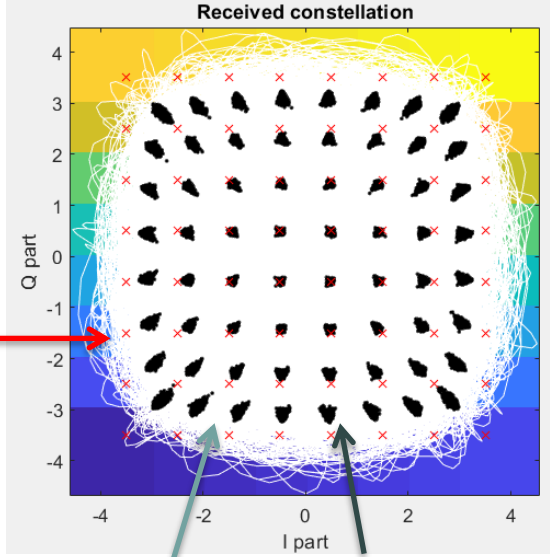
$$ACPR_{low} = \frac{P_{out}}{P_{\text{low}}}$$

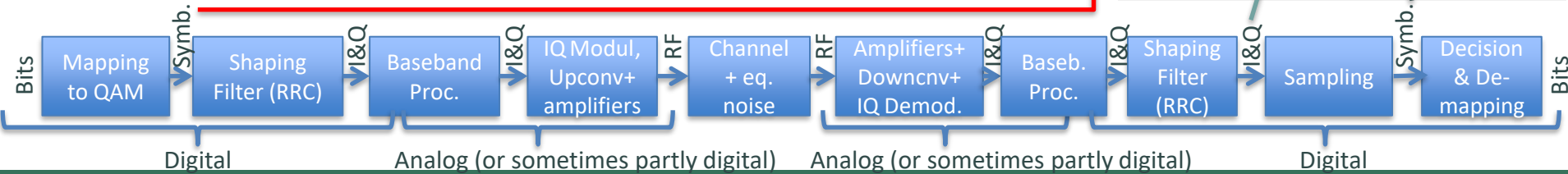
$$ACPR_{up} = \frac{P_{out}}{P_{\text{up}}}$$

ACPR measures the signal spilling into adjacent channels, caused by nonlinearities and spurious emissions.
It represents the “degradation” by the transmitter on the adjacent channels.
NOTE: The position and bandwidth of adjacent channels must be specified together the ACPR requirements

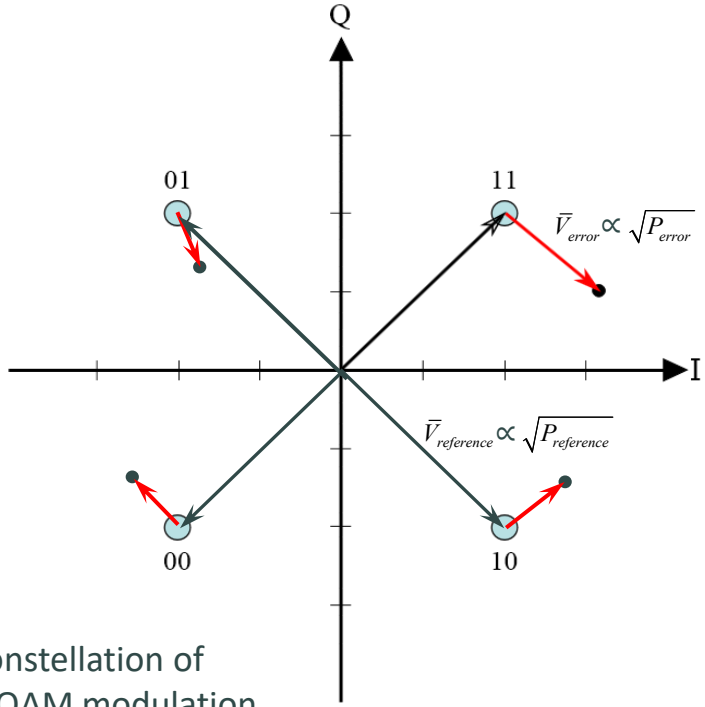
Degradation of Digital Signals

- In digital communications, information is embedded into symbols, each in a predefined complex position (constellation diagram)
- Distortions and noise and interferences… all induce a displacement between the transmitted symbol and the actual received symbol
- If the received symbol is closer to another allowed position in the constellation, the decision and the returned information is wrong → Errors in the bit stream!
- Such displacement is an indicator of the overall degradation of the signal





Error Vector Magnitude (EVM)



Constellation of
4-QAM modulation

Given the error vectors between the ideal constellation and the degraded signal, EVM is defined as follows:

$$EVM_{\%} = 100 \cdot \sqrt{\frac{P_{error}}{P_{reference}}}$$

$$EVM_{dB} = 20 \log_{10} \sqrt{\frac{P_{error}}{P_{reference}}}$$

with:

P_{error}=average power of the error vectors

$P_{reference}$ = average power of the reference vectors

Non-linear systems with memory

- In non-linear systems, memory means that the response to an excitation at a given time *t* depends also on the time history prior to *t*.

$$v_o(t) = a_1 v_i(t) + a_{30} v_i^3(t-\tau) + a_{21} v_i^2(t-\tau_2) v(t-\tau_3) + a_{12} v_i\ (t-\tau_4) v^2(t-\tau_5) + \ldots$$

- A simple polynomial model is usually no longer adequate to estimate the response
- Memory effects are detected as **AM to PM conversion**

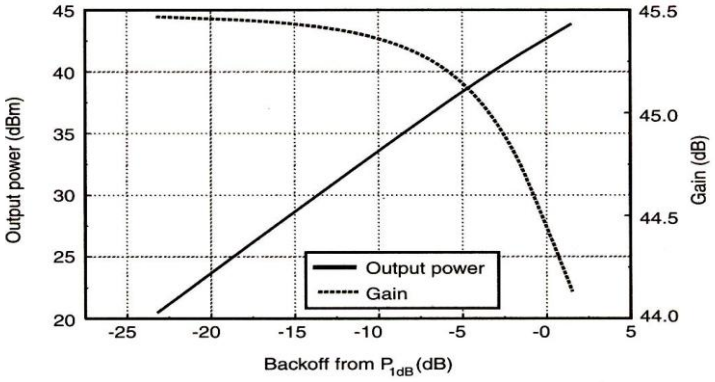
an input sinusoid:

$$s_{RF}(t) = V_M(t)\cos(\omega_0 t)$$

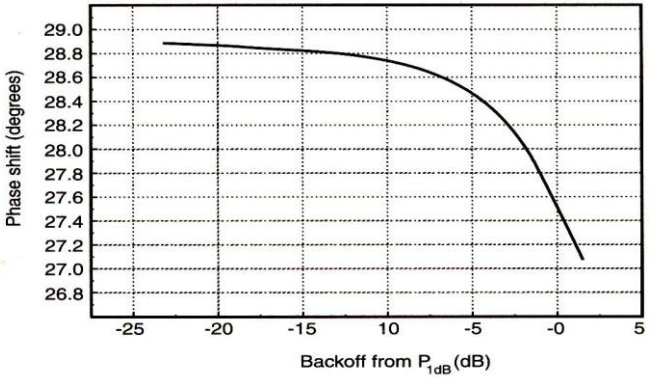
is output as:

$$s_{out}(t) \equiv f(V_M(t))\cos(\omega_0 t + \phi_0) \qquad \phi_o(\omega_0) \equiv g(P_{out})$$

AM-AM Conversion (based on input amplitude, output is further amplitude-modulated (compression))

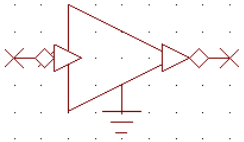


AM-PM Conversion (based on input amplitude, output is further phase-modulated)



Behavioral Models of Amplifiers

AMP_B2
ID=A1
GAIN=10
P1DB=10
IP3=
IP2=
MEASREF=
OSAT=
NF=3
NOISE=Auto
RFIFRQ=



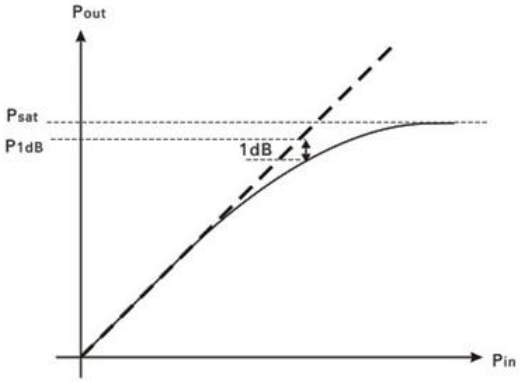
All CAD tools (both systems and circuital): various behavioral model for amplifiers.

- noise (through the Noise Figure)
- and non-linearity (by means P_{1dB} and IP_3).

The most sophisticated models represent the effect of output saturation (above P1db) and the weak memory effect (AM-PM conversion).

VSS Model of the real amplifier

- The nonlinearity is approximated using a fifth-order polynomial
- P_{1dB}, IP_2 and IP_3 are available
- Saturation power Psat can also be specified
- AM-PM conversion is not implemented in the basic model



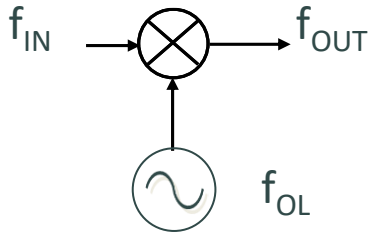
Non-Linear Behavior of Mixers

Mixers: non-linear by definition!

Minimum polynomial model: $v_o=(v_i)^2$ with $v_i=V_{RF}\cos(\omega_{IN}t)+V_{OL}\cos(\omega_{OL}t)$

But: real mixers also exhibit non-linear distortion

Large RF input signal→ the spectrum of the converted signal at output is distorted (as in amplifiers).



A polynomial model can be used for characterization, like in memoryless PA:

$$v_o = a_0 + a_1 v_i + \boxed{a_2 v_i^2} + a_3 v_i^3 + a_4 v_i^4 \ldots.$$

However: a pure 3-order characteristic $a_1v_i+a_3v_i^3$ does not produce distortion at $f_{IN}\pm f_{OL}$

In general: odd orders create $|2f_{IN}\pm f_{OL}|$, $|f_{IN}\pm 2f_{OL}|$, $|f_{IN}\pm 4f_{OL}|$ $|3f_{IN}\pm 2f_{OL}|$…→easily filtered out

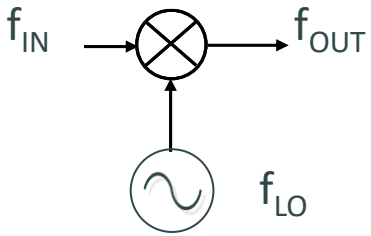
The fundamental mixing products around $f_{IN}\pm f_{OL}$ come from the even terms!

a_4, a_6 … are the main reason for non-linear distortion in mixers!

Characterization of Mixers in Large Signal Regime

Characterization is made with the same parameters used for PA (**P_{1dB} and IP_3**), with some differences.

- The input/output "useful components" used to evaluate power levels are at different frequencies (f_{RF} at input and $f_{(RF\pm LO)}$ at output)
- These parameters depend on the power of local oscillator (P_{LO}): typically, there is a range of P_{LO} in which P_{1dB} and IP_3 remain constant. (usually the LO must keep the mixer in "saturation", e.g. in the relevant non-linear region where a_2 is high)
- Usually the value of P_{1dB} and IP_3 in a mixer are **referred at the input** (and not at output as in PA):

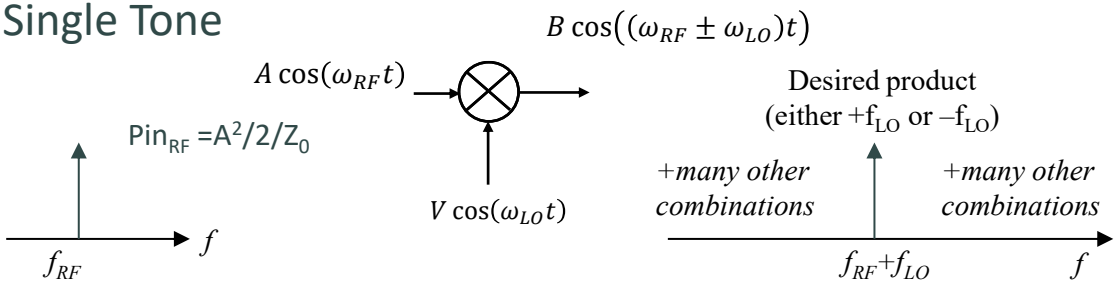


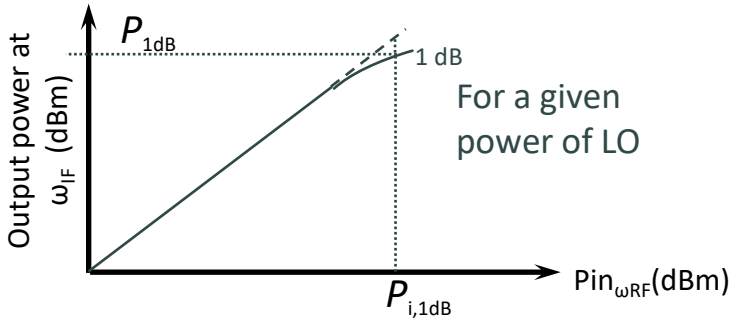
$$P_{i,1db,dBm} = P_{1dB,dBm} + 1dB - G_{dB},$$
$$IIP_{3,dBm} = IP_{3,dBm} - G_{dB}$$

GdB: the nominal gain (i.e. opposite of the conversion loss in dB), also known as "linear" gain or "small-signal" gain

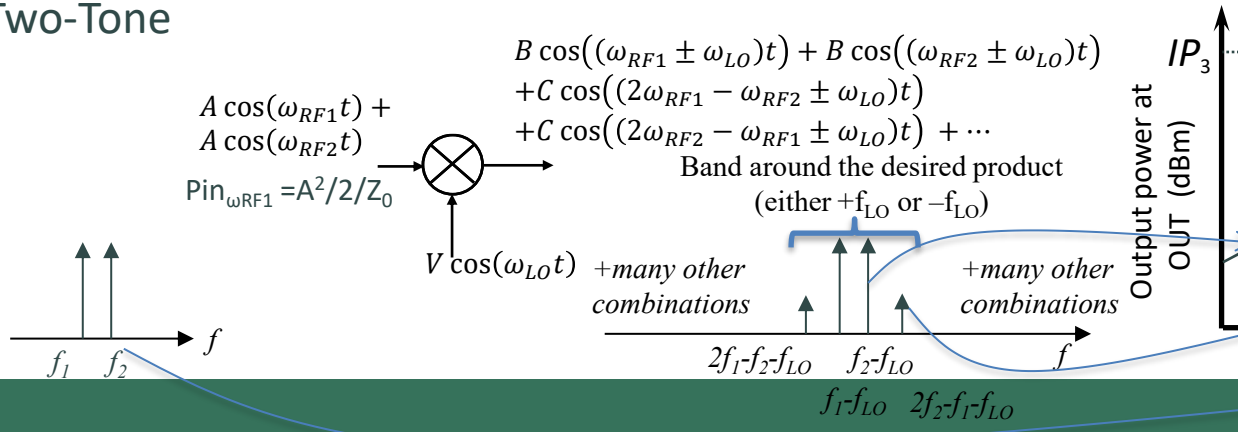
P1dB and IIP3 in Mixers

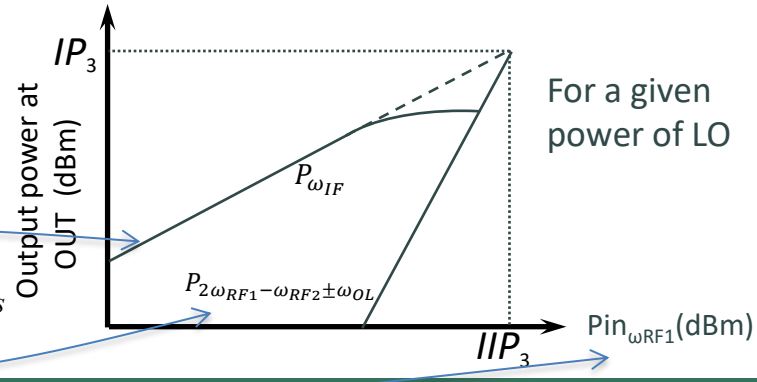
Single Tone



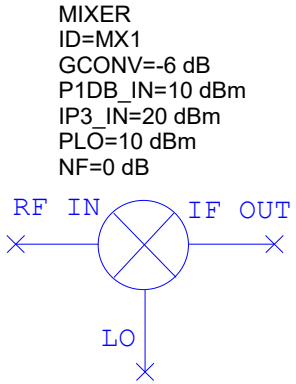


Two-Tone





Behavioral Models of Mixers and Harmonic Mixing



Main parameters:

P1dB (1 dB Compression Point referred to RF IN)

IP3 or TOI (3rd order intercept power referred to RF IN)

GCONV (small signal gain, from f_{RF} to the desired f_{IF} tone)

PLO (LO power for specified conversion gain)

TSSB or TDSB or NF or NF_{IEEE} (equivalent noise temperatures)

The non-linear characteristic of a real mixer potentially creates all components:

$$f_{out} = |\pm m \cdot f_{RF} \pm n \cdot f_{LO}| \qquad n, m = 1,2,\ldots.$$

Beware of undesired mixing products (i.e. spurious oscillator frequencies, local oscillator residuals, cross talk…) which may end up in your IF signal!

In practice, only |m|+|n|<≈5 are usually relevant: think about filtering and careful system design!

Sub-harmonic mixers instead work with one of the harmonics of f_{LO} to generate the desired frequency conversion:

$f_{out} = f_{RF} \pm n \cdot f_{LO}$ $\quad$ n=3 ÷ 5 $\quad$ They require a lower frequency LO, but the conversion loss is usually higher